

Blood Flow in Microvascular Networks

Coursework 2: Individual Problem Sheet

Exercise 1

The human body is composed of a continuous hierarchy of vessels, with topologies and geometries varying depending on the tissues. There are three types of microvessels: arterioles, capillaries, and venules. Capillaries are fed by arterioles and drained by venules. From Topic 4 lecture notes, it was established that vessels can be classified according to their diameter: arterioles and venules are in the range of approximately $100\mu\text{m}$ to 1 mm (so radius between $50\mu\text{m}$ to $100\mu\text{m}$), whereas capillaries are in the range of approximately $10\mu\text{m}$ to $100\mu\text{m}$ (so radius between $5\mu\text{m}$ to $50\mu\text{m}$).

Secondly, it can be assumed that Poiseuille's Law $Q = \frac{\pi R^4 \Delta P}{8\mu L}$ holds, for Q being the flow rate, ΔP the pressure difference, L the length of the vessel, R its radius, and μ the blood viscosity which is constant (equal to $0.004\text{ Pa}\cdot\text{s}$). In Poiseuille's Law, the dominant term is the radius raised to the power of 4. This means that the flow rate Q has a heavy dependence on the R^4 term, and so as the radius of the vessel increases, the flow rate also increases exponentially. For instance, a microvessel that doubles in radius would have 16 times higher fluid flow.

From this information, the following sketch can be obtained:

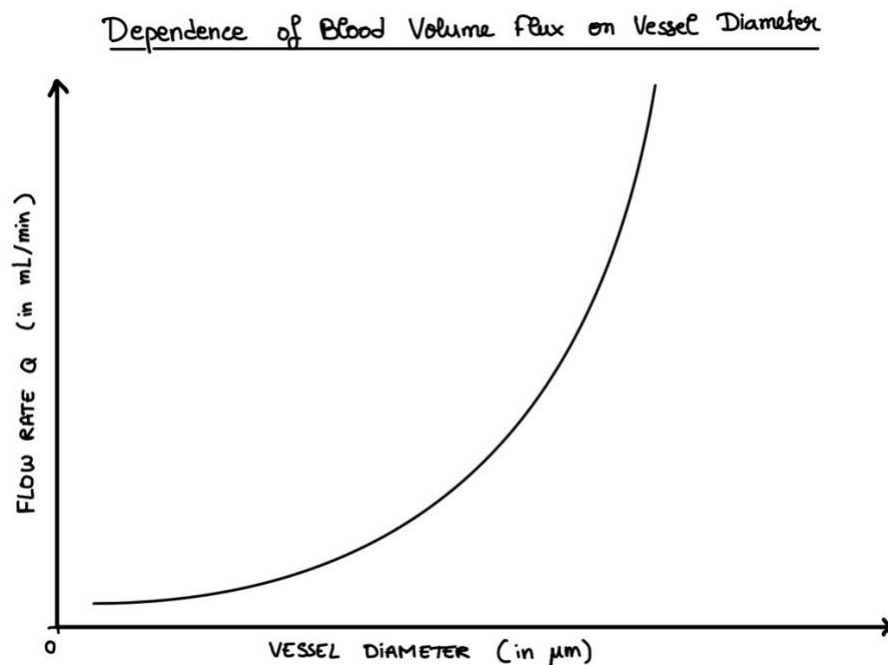


Figure 1: Sketch showing the dependence of blood volume flux on the vessel diameter

Exercise 2

- a) First, the Poiseuille's Law links the flow rate Q , the radius of the vessel R , the pressure difference ΔP , the blood viscosity μ and the length of the vessel L , such that:

$$Q = \frac{\pi R^4 \Delta P}{8\mu L}$$

Considering a parent vessel which bifurcates into two daughter segments, and using Figure 1a), the pressures p_i at nodes $i = 1,2,3,4$ and fluxes Q_j in segments $j = 1,2,3$ can be denoted such that:

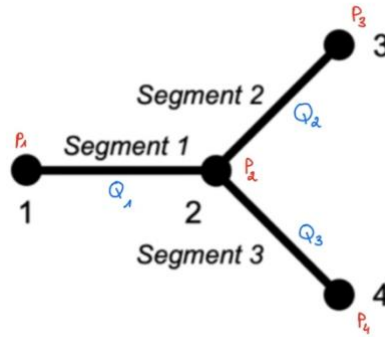


Figure 2: Annotated Schematic of a Vessel Bifurcation, taken from Figure 1a) of assignment brief.

Where p_1 describes the pressure at node 1, p_2 the pressure at node 2, p_3 the pressure at node 3, and p_4 the pressure at node 4. Likewise, the flux flowing through segment 1 is described by Q_1 , the flux flowing through segment 2 by Q_2 , and the flux flowing through segment 3 by Q_3 .

Assuming that each segment has a different diameter, the radius of the vessel will change depending on the segment. Similarly, each segment has a different length. Thus, segment 1 will have a radius of R_1 and length of L_1 , segment 2 will have a radius of R_2 and length of L_2 , and segment 3 will have a radius of R_3 and length of L_3 . Finally, knowing that the viscosity remains constant, the following expressions can be obtained from Poiseuille's Law for each flux:

$$Q_1 = \frac{\pi R_1^4 (p_1 - p_2)}{8\mu L_1}$$

$$Q_2 = \frac{\pi R_2^4 (p_2 - p_3)}{8\mu L_2}$$

$$Q_3 = \frac{\pi R_3^4 (p_2 - p_4)}{8\mu L_3}$$

- b) The studied fluid is blood, which is an incompressible fluid (its mass per unit volume is constant). Thus, the conservation of mass principle can be applied, which implies conservation of volume. This means that the volume flux Q_1 that goes in the vessel (through segment 1) must equal the fluxes Q_2 and Q_3 that go out of the vessel (through segments 2 & 3), and that the three volume fluxes can therefore be related by the following expression:

$$Q_1 = Q_2 + Q_3$$

- c) Assuming that the pressures at the boundary nodes (1,3,4) are known, the pressure at the interior node p_2 can be found using:

$$Q_1 = Q_2 + Q_3$$

$$\frac{\pi R_1^4 (p_1 - p_2)}{8\mu L_1} = \frac{\pi R_2^4 (p_2 - p_3)}{8\mu L_2} + \frac{\pi R_3^4 (p_2 - p_4)}{8\mu L_3}$$

Where isolating the variable p_2 is now required. First, multiply both sides by $8\mu L_1$:

$$\pi R_1^4 (p_1 - p_2) = \frac{\pi R_2^4 (p_2 - p_3) 8\mu L_1}{8\mu L_2} + \frac{\pi R_3^4 (p_2 - p_4) 8\mu L_1}{8\mu L_3}$$

$$\pi R_1^4 (p_1 - p_2) = \frac{\pi R_2^4 (p_2 - p_3) L_1}{L_2} + \frac{\pi R_3^4 (p_2 - p_4) L_1}{L_3}$$

Dividing everything by π :

$$R_1^4 (p_1 - p_2) = \frac{R_2^4 (p_2 - p_3) L_1}{L_2} + \frac{R_3^4 (p_2 - p_4) L_1}{L_3}$$

$$R_1^4 (p_1 - p_2) = \frac{R_2^4 (p_2 - p_3) L_1 L_3 + R_3^4 (p_2 - p_4) L_1 L_2}{L_2 L_3}$$

$$R_1^4 (p_1 - p_2) L_2 L_3 = R_2^4 (p_2 - p_3) L_1 L_3 + R_3^4 (p_2 - p_4) L_1 L_2$$

Expand everything:

$$L_2 L_3 R_1^4 p_1 - L_2 L_3 R_1^4 p_2 = L_1 L_3 R_2^4 p_2 - L_1 L_3 R_2^4 p_3 + L_1 L_2 R_3^4 p_2 - L_1 L_2 R_3^4 p_4$$

$$L_1 L_3 R_2^4 p_2 + L_1 L_2 R_3^4 p_2 + L_2 L_3 R_1^4 p_2 = L_2 L_3 R_1^4 p_1 + L_1 L_3 R_2^4 p_3 + L_1 L_2 R_3^4 p_4$$

$$p_2 (L_2 L_3 R_1^4 + L_1 L_3 R_2^4 + L_1 L_2 R_3^4) = L_2 L_3 R_1^4 p_1 + L_1 L_3 R_2^4 p_3 + L_1 L_2 R_3^4 p_4$$

Finally:

$$p_2 = \frac{L_2 L_3 R_1^4 p_1 + L_1 L_3 R_2^4 p_3 + L_1 L_2 R_3^4 p_4}{L_2 L_3 R_1^4 + L_1 L_3 R_2^4 + L_1 L_2 R_3^4}$$

Exercise 3

- a) Based on Poiseuille's Law for flow rate, for a decrease of pressure between the two boundaries, there is an increase in the vessel's radius (thus diameter). Since each segment has a different diameter, we can expect this increase to be gradual.

As seen in exercise 1 and according to lecture notes, vessels with a diameter in the range of 10 to 100 microns are capillaries (and so their radius is between $5\mu\text{m}$ to $50\mu\text{m}$). Thus, it would be logical to associate the length of a capillary to each segment. It can be found that a capillary usually

measures around 0.5 mm^{12} . Furthermore, from the lecture notes, it can be assumed that blood viscosity is approximately 0.004 Pa s .

The decrease in blood pressure usually occurs in capillaries when the arterioles dilate³. From exercise 1, it was established that the radius of an arteriole ranges from $50 \mu\text{m}$ to $100 \mu\text{m}$. Therefore, we could assume that these capillaries are close to arterioles, and so taking radii close to $50 \mu\text{m}$ (maximum radius of capillary and minimum radius of arteriole) can be reasonable. Finally, since the segments only measure half a millimeter, the change of radius between each vessel should be small, for instance it shouldn't be larger than around 1 to $2 \mu\text{m}$.

In summary, from these findings or assumptions, we have:

Parameter	Physiologically reasonable values
Blood viscosity μ	0.004 Pa s
Vessel Length L	0.5 mm
Radius of segment 1 R_1	$42 \mu\text{m}$
Radius of segment 2 R_2	$44 \mu\text{m}$
Radius of segment 3 R_3	$46 \mu\text{m}$
Radius of segment 4 R_4	$43 \mu\text{m}$
Radius of segment 5 R_5	$45 \mu\text{m}$
Radius of segment 6 R_6	$47 \mu\text{m}$
Radius of segment 7 R_7	$49 \mu\text{m}$
Radius of segment 8 R_8	$50 \mu\text{m}$

- b) Like Figure 2 in exercise 2, the “ladder” network can be labelled such that p_i represents the pressure at node i and Q_j the flux in segment j for $i=1,2,3,4,5,6,7,8$ and $j=1,2,3,4,5,6,7,8$:

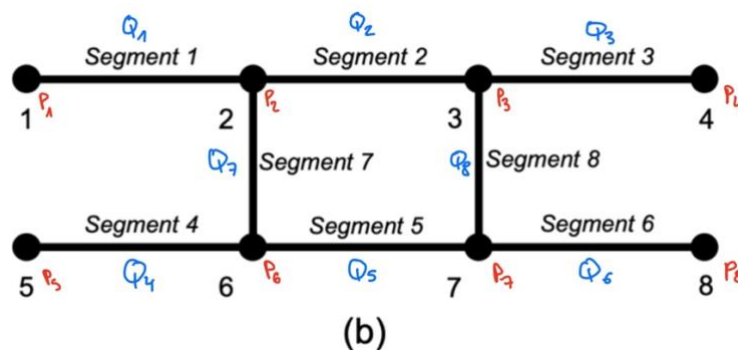


Figure 3: Annotated Schematic of “Ladder” Network, taken from Figure 1b) of assignment brief

¹ MEFANET (2022) *Capillary*, WikiLectures. Available at: <https://www.wikilectures.eu/w/Capillary#:~:text=The%20microcirculation%20is%20a%20very,ranges%20from%200.25%2D1%20mm> (Accessed: 12 March 2024)

² Fractal Blood Vessel (no date) *Fractal Foundation Online Course - Chapter 1 - fractals in nature*. Available at: <https://fractalfoundation.org/OFC/OFC-1-3.html#:~:text=And%20according%20to%20another%20study,just%20over%20half%20a%20millimeter.&text=Using%20these%20numbers%2C%20we%20can,of%20capillaries%20in%20a%20human.> (Accessed: 12 March 2024).

³ Why does blood pressure drop in the capillaries? (no date) byju. Available at: <https://byjus.com/question-answer/why-does-blood-pressure-drop-in-the-capillaries/> (Accessed: 12 March 2024).

It is known that the pressures at nodes 1 & 5 are equal to 40 mmHg, and that the pressures at nodes 4 & 8 are 15 mmHg. As seen in Figure 3, nodes 1, 5, 4 and 8 are the boundary pressures. We are asked to derive 4 equations for the unknown pressures, i.e. pressures at nodes 2, 3, 6 and 7. This can be done using the same approach as the bifurcation network in exercise 2.

First, using Poiseuille's Law and assuming each segment has a different radius R_i but equal length L and viscosity μ , the fluxes in each segment can be expressed such that:

$$Q_1 = \frac{\pi R_1^4 (p_1 - p_2)}{8\mu L}$$

$$Q_2 = \frac{\pi R_2^4 (p_2 - p_3)}{8\mu L}$$

$$Q_3 = \frac{\pi R_3^4 (p_3 - p_4)}{8\mu L}$$

$$Q_4 = \frac{\pi R_4^4 (p_5 - p_6)}{8\mu L}$$

$$Q_5 = \frac{\pi R_5^4 (p_6 - p_7)}{8\mu L}$$

$$Q_6 = \frac{\pi R_6^4 (p_7 - p_8)}{8\mu L}$$

$$Q_7 = \frac{\pi R_7^4 (p_2 - p_6)}{8\mu L}$$

$$Q_8 = \frac{\pi R_8^4 (p_3 - p_7)}{8\mu L}$$

Then, assuming the blood flows in only one direction – from left to right, and from top to bottom – the following 4 equations can be obtained:

$$Q_1 = Q_2 + Q_7 \quad (1)$$

$$Q_2 = Q_3 + Q_8 \quad (2)$$

$$Q_5 = Q_4 + Q_7 \quad (3)$$

$$Q_6 = Q_5 + Q_8 \quad (4)$$

For equation (1), the known boundary pressure is p_1 :

$$\frac{\pi R_1^4 (p_1 - p_2)}{8\mu L} = \frac{\pi R_2^4 (p_2 - p_3)}{8\mu L} + \frac{\pi R_7^4 (p_2 - p_6)}{8\mu L}$$

Multiply by $8\mu L$ and divide by π :

$$\begin{aligned} R_1^4 (p_1 - p_2) &= R_2^4 (p_2 - p_3) + R_7^4 (p_2 - p_6) \\ R_1^4 p_1 - R_1^4 p_2 &= R_2^4 (p_2 - p_3) + R_7^4 (p_2 - p_6) \\ p_1 &= \frac{R_2^4 (p_2 - p_3) + R_7^4 (p_2 - p_6) + R_1^4 p_2}{R_1^4} \\ p_1 &= \frac{p_2 (R_1^4 + R_2^4 + R_7^4) - p_3 R_2^4 - p_6 R_7^4}{R_1^4} \end{aligned}$$

For equation (2), the known boundary pressure is p_4 :

$$\begin{aligned}
 Q_2 &= Q_3 + Q_8 \\
 \frac{\pi R_2^4 (p_2 - p_3)}{8\mu L} &= \frac{\pi R_3^4 (p_3 - p_4)}{8\mu L} + \frac{\pi R_8^4 (p_3 - p_7)}{8\mu L} \\
 R_2^4 (p_2 - p_3) &= R_3^4 p_3 - R_3^4 p_4 + R_8^4 (p_3 - p_7) \\
 p_4 &= \frac{R_3^4 p_3 + R_8^4 (p_3 - p_7) - R_2^4 (p_2 - p_3)}{R_3^4} \\
 p_4 &= \frac{p_3 (R_3^4 + R_8^4 + R_2^4) - p_7 R_8^4 - p_2 R_2^4}{R_3^4}
 \end{aligned}$$

For equation (3), the known boundary pressure is p_5 :

$$\begin{aligned}
 Q_5 &= Q_4 + Q_7 \\
 \frac{\pi R_5^4 (p_6 - p_7)}{8\mu L} &= \frac{\pi R_4^4 (p_5 - p_6)}{8\mu L} + \frac{\pi R_7^4 (p_2 - p_6)}{8\mu L} \\
 R_5^4 (p_6 - p_7) &= R_4^4 p_5 - R_4^4 p_6 + R_7^4 (p_2 - p_6) \\
 p_5 &= \frac{R_5^4 (p_6 - p_7) + R_4^4 p_6 - R_7^4 (p_2 - p_6)}{R_4^4} \\
 p_5 &= \frac{p_6 (R_5^4 + R_4^4 + R_7^4) - p_7 R_5^4 - p_2 R_7^4}{R_4^4}
 \end{aligned}$$

For equation (4), the known boundary pressure is p_8 :

$$\begin{aligned}
 Q_6 &= Q_5 + Q_8 \\
 \frac{\pi R_6^4 (p_7 - p_8)}{8\mu L} &= \frac{\pi R_5^4 (p_6 - p_7)}{8\mu L} + \frac{\pi R_8^4 (p_3 - p_7)}{8\mu L} \\
 R_6^4 p_7 - R_6^4 p_8 &= R_5^4 (p_6 - p_7) + R_8^4 (p_3 - p_7) \\
 p_8 &= \frac{R_6^4 p_7 - R_5^4 (p_6 - p_7) - R_8^4 (p_3 - p_7)}{R_6^4} \\
 p_8 &= \frac{p_7 (R_6^4 + R_5^4 + R_8^4) - p_6 R_5^4 - p_3 R_8^4}{R_6^4}
 \end{aligned}$$

- c) Solving for the unknown pressures can be done using matrix methods for linear systems, where $\mathbf{Ax} = \mathbf{b}$, such that:

$$A = \begin{pmatrix} \frac{R_1^4 + R_2^4 + R_7^4}{R_1^4} & -\frac{R_2^4}{R_1^4} & -\frac{R_7^4}{R_1^4} & 0 \\ -\frac{R_2^4}{R_3^4} & \frac{R_2^4 + R_3^4 + R_8^4}{R_3^4} & 0 & -\frac{R_8^4}{R_3^4} \\ -\frac{R_7^4}{R_4^4} & 0 & \frac{R_4^4 + R_5^4 + R_7^4}{R_4^4} & -\frac{R_5^4}{R_4^4} \\ 0 & -\frac{R_8^4}{R_6^4} & -\frac{R_5^4}{R_6^4} & \frac{R_5^4 + R_6^4 + R_8^4}{R_6^4} \end{pmatrix}$$

$$x = \begin{pmatrix} p_2 \\ p_3 \\ p_6 \\ p_7 \end{pmatrix}$$

$$b = \begin{pmatrix} p_1 \\ p_4 \\ p_5 \\ p_8 \end{pmatrix}$$

Applying numerical values to A gives:

$$A = \begin{pmatrix} \frac{42^4 + 44^4 + 49^4}{42^4} & -\frac{44^4}{42^4} & -\frac{49^4}{42^4} & 0 \\ -\frac{44^4}{46^4} & \frac{44^4 + 46^4 + 50^4}{46^4} & 0 & -\frac{50^4}{46^4} \\ -\frac{49^4}{43^4} & 0 & \frac{43^4 + 45^4 + 49^4}{43^4} & -\frac{45^4}{43^4} \\ 0 & -\frac{50^4}{47^4} & -\frac{45^4}{47^4} & \frac{45^4 + 47^4 + 50^4}{47^4} \end{pmatrix}$$

$$A \approx \begin{pmatrix} 4.06 & -1.20 & -1.85 & 0 \\ -0.84 & 3.23 & 0 & -1.40 \\ -1.69 & 0 & 3.88 & -1.20 \\ 0 & -1.28 & -0.84 & 3.12 \end{pmatrix}$$

Applying numerical values to b :

$$b = \begin{pmatrix} 40 \\ 15 \\ 40 \\ 15 \end{pmatrix}$$

Thus:

$$\begin{pmatrix} 4.06 & -1.20 & -1.85 & 0 \\ -0.84 & 3.23 & 0 & -1.40 \\ -1.69 & 0 & 3.88 & -1.20 \\ 0 & -1.28 & -0.84 & 3.12 \end{pmatrix} \begin{pmatrix} p_2 \\ p_3 \\ p_6 \\ p_7 \end{pmatrix} = \begin{pmatrix} 40 \\ 15 \\ 40 \\ 15 \end{pmatrix}$$

Solving it such that $x = A^{-1}b$ gives:

$$\begin{pmatrix} p_2 \\ p_3 \\ p_6 \\ p_7 \end{pmatrix} = \begin{pmatrix} 4.06 & -1.20 & -1.85 & 0 \\ -0.84 & 3.23 & 0 & -1.40 \\ -1.69 & 0 & 3.88 & -1.20 \\ 0 & -1.28 & -0.84 & 3.12 \end{pmatrix}^{-1} \begin{pmatrix} 40 \\ 15 \\ 40 \\ 15 \end{pmatrix}$$

$$\begin{pmatrix} p_2 \\ p_3 \\ p_6 \\ p_7 \end{pmatrix} = \begin{pmatrix} 30.12006476 \\ 22.00367609 \\ 30.22878142 \\ 21.97335955 \end{pmatrix} \approx \begin{pmatrix} 30 \\ 22 \\ 30 \\ 22 \end{pmatrix}$$

Having found all the pressures, they can be summarized in the table below:

Node	Pressure in <i>mmHg</i>	Pressure in <i>Pa</i>
p_1	40	≈ 5333
p_2	30	≈ 4000
p_3	22	≈ 2933
p_4	15	≈ 2000
p_5	40	≈ 5333

p_6	30	≈ 4000
p_7	22	≈ 2933
p_8	15	≈ 2000

Those values can now be used to calculate the segment fluxes:

$$Q_1 = \frac{\pi \times (42 \times 10^{-6})^4 (5333 - 4000)}{8 \times 0.004 \times 0.5 \times 10^{-3}} \approx 8.14 \times 10^{-10} \frac{m^3}{s} = 0.0488 \text{ mL/min}$$

$$Q_2 = \frac{\pi \times (44 \times 10^{-6})^4 (4000 - 2933)}{8 \times 0.004 \times 0.5 \times 10^{-3}} \approx 7.85 \times 10^{-10} \frac{m^3}{s} = 0.0471 \text{ mL/min}$$

$$Q_3 = \frac{\pi \times (46 \times 10^{-6})^4 (2933 - 2000)}{8 \times 0.004 \times 0.5 \times 10^{-3}} \approx 8.20 \times 10^{-10} \frac{m^3}{s} = 0.0492 \text{ mL/min}$$

$$Q_4 = \frac{\pi \times (43 \times 10^{-6})^4 (5333 - 4000)}{8 \times 0.004 \times 0.5 \times 10^{-3}} \approx 8.95 \times 10^{-10} \frac{m^3}{s} = 0.0537 \text{ mL/min}$$

$$Q_5 = \frac{\pi \times (45 \times 10^{-6})^4 (4000 - 2933)}{8 \times 0.004 \times 0.5 \times 10^{-3}} \approx 8.59 \times 10^{-10} \frac{m^3}{s} = 0.0515 \text{ mL/min}$$

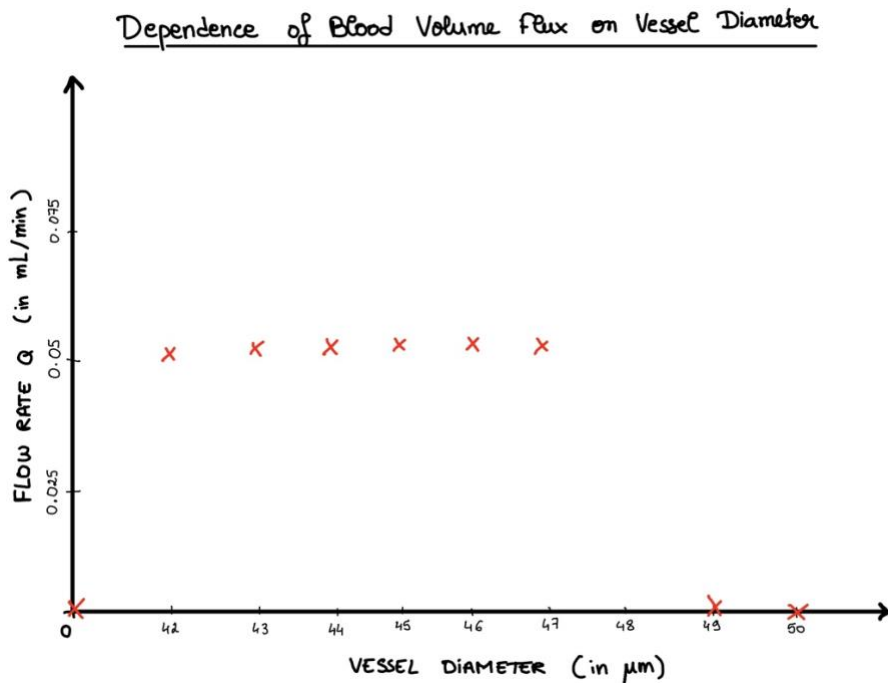
$$Q_6 = \frac{\pi \times (47 \times 10^{-6})^4 (2933 - 2000)}{8 \times 0.004 \times 0.5 \times 10^{-3}} \approx 8.94 \times 10^{-10} \frac{m^3}{s} = 0.0536 \text{ mL/min}$$

$$Q_7 = \frac{\pi \times (49 \times 10^{-6})^4 (4000 - 4000)}{8 \times 0.004 \times 0.5 \times 10^{-3}} = 0 \text{ mL/min}$$

$$Q_8 = \frac{\pi \times (50 \times 10^{-6})^4 (2933 - 2933)}{8 \times 0.004 \times 0.5 \times 10^{-3}} = 0 \text{ mL/min}$$

The fluxes in segments 1 to 6 are of approximately 0.05 mL/min, and that there are no fluxes in segments 7 and 8 (because there is no pressure gradient). Note that putting these values (rounded to 0.05 mL/min) in the four equations derived in part b) allows us to check that they make sense.

From this information, the following sketch can be obtained:



From the reasoning of exercise 1, it was expected to see an exponential increase in segment flux as the diameter of the vessel increases, due to the dependence on the radius raised to the power of 4.

However, the values obtained show that this is not the case, as there is a decrease in flow rate even if the diameter is larger. This indicates that the blood flux in each segment does not only depend on the vessel diameter, and that at certain points, even if the diameter of the vessel is increasing, other factors such as the pressure gradient, viscosity and length play a more significant role in dictating the flow rate.

For instance, in the segments with no pressure gradient - such as segments 7 and 8 - there was no blood flow, despite them having a larger diameter than all the other segments. This contradicts the assumption made in exercise 1. In those cases, the radius raised to the power of 4 had no direct correlation or impact on flow rate, as it is a pressure difference that is needed to drive flow in the first place.

Furthermore, as mentioned previously, factors such as viscosity and vessel length were considered as constant in our calculations. However, these factors vary in the human body. Some vessels can be longer or shorter than others, which in turn would also affect the flow rate (based on Poiseuille's law).

Overall, these assumptions and findings indicate that the sketch assumed in exercise 1 was imprecise.